

Lok Swar: Comprehensive Architecture, APIs, AI Models & Software Stack

A deep-dive technical blueprint covering every feature, internal/external API, artificial intelligence pipeline, data fusion engine, and zero-literacy interface built for **Lok Swar**.

1. Executive Overview & The Problem Solved

Lok Swar ("The Voice of the People") is a decentralized, multimodal civic grievance, participatory democracy, and public infrastructure audit platform tailored for rural and semi-urban India.

🔥 The Rural Barrier

Over 280 million rural citizens in India face illiteracy or digital hesitation. Existing portals (CPGRAMS, district grievance forms) require typed English/Hindi text and complex dropdown trees, excluding ground reality.

💡 The Lok Swar Paradigm

Zero-Literacy Voice-First UX: Tap anywhere to listen. Speak in raw regional dialect (Bhojpuri, Odia, Bengali, Hindi). Automatic satellite GPS sync. Multi-sensor verification against ghost work and corruption.

2. Comprehensive Feature Breakdown

A. Citizen Portal Features (`citizen.html`)

- **Every-Click Spoken Guidance (Zero-Literacy Engine):** Illiterate citizens tapping anywhere on the screen trigger immediate conversational speech orientation in their native tongue without needing to read instructions.
- **Automatic GPS & Telemetry Sync:** Completely eliminates manual "Refresh GPS" buttons. Automatically acquires satellite coordinates on load, reverse geocodes locality down to Ward/Gram Panchayat, and maintains live movement telemetry via `navigator.geolocation.watchPosition`.
- **Location-Driven UI Language Adaptation:** Automatically detects geographic state/district from GPS/IP and transforms the entire interface (titles, cards, badges, spoken speech) to the local language:

Bihar → Bhojpuri

Odisha → Odia

West Bengal → Bengali

North India → Hindi

- **Continuous Search Bar Auto-Translation:** As citizens speak or type in their local dialect, a dual-layer NMT pipeline continuously translates colloquial sentences into polished Administrative English in real time directly inside the search bar.
- **Multimodal Grievance Submission:** Supports simultaneous submission of raw voice audio recordings (base64 WAV/WebM), camera-captured photo evidence, GPS coordinates, and Aadhaar identity verification.
- **Democratic Civic Upvoting & Crowdsourced Verification:** Citizens living in the same Gram Panchayat can view nearby grievances on an interactive Leaflet map and endorse ("Upvote") them, boosting priority without filing redundant duplicate reports.
- **Profile & DigiLocker Identity Integration:** Seamless Aadhaar/Voter ID verification fetching simulated verified credentials while preserving citizen privacy through SHA-256 mobile salting.

B. Executive & Administrative Dashboard (`admin.html`)

- **Multi-Dimensional Spatial Heatmaps:** Real-time GIS rendering with Leaflet.js showcasing high-density infrastructure breakdown clusters using spatial clustering algorithms.
- **Autonomous UAV Drone Surveillance Dispatch:** Administrative dispatch of drone recon missions to verify high-urgency road washouts, electrical hazards, or school roof collapses before allocating state budget.
- **Algorithmic Treasury Budget Allocation:** Interactive budget ledger matching civic grievances directly against official state and central funding schemes (e.g. *Jal Jeevan Mission, PMGSY, SDRF Disaster Relief, 5T Transformation*).
- **Multi-Sensor Data Fusion Center:** Correlates citizen reports against independent ground sensors, weather satellite feeds, and satellite SAR to detect false or exaggerated claims.
- **Contractor Adjudication & Accountability Matrix:** Automated tracking of maintenance SLAs, breach penalties, and public contractor resolution logs.

3. Artificial Intelligence, NLP & Audio Models

Lok Swar utilizes a multi-tier AI pipeline engineered to handle dialect variation, noisy village audio, and official administrative synthesis.

AI ENGINE / TOOL	UNDERLYING MODEL / TECH	INPUT	OUTPUT	FAIL-SAFE FALLBACK
Automatic Speech Recognition (ASR)	Google Speech API & Web Speech API (<code>SpeechRecognition</code>)	Raw citizen voice audio (Odia, Bhojpuri, Hindi, Bengali)	UTF-8 Devanagari / Odia text transcript	Server Python <code>speech_recognition</code> with PyAudio / soundfile

AI ENGINE / TOOL	UNDERLYING MODEL / TECH	INPUT	OUTPUT	FAIL-SAFE FALLBACK
Colloquial Dialect Normalizer	Rule-Based Regex & Tokenized Grammatical Transducers	Raw dialect phrases (e.g. "पुलवा बह गइल बा", "लाइन नइखे")	Normalized concept representation	Exact phonetic match lexicon
Neural Machine Translation (NMT)	Google Translate GTX + MyMemory Multi-Lingual API	Colloquial regional speech text	Concise Administrative English for executive triage	Domain-specific infrastructure bilingual phrasebook
NLP Automatic Categorization	Multilingual Keyword Token Matcher + Semantic Scorer	Combined raw speech + English translated text	1 of 7 Civic Categories (Power, Roads, Water, School, PHC, Drain, Canal)	Default to Roads & Connectivity
Urgency & Priority Scoring Engine	Dynamic Risk Heuristic with Hazard Boosts	Incident category, emergency keywords (<i>collapse, flood, blast</i>), verification status	Priority Urgency Score (0.0 to 100.0)	Default category baseline (e.g. PHC: 95.0, Roads: 92.5)
Government Scheme Matcher	Deterministic Scheme Knowledge Base	Incident category and administrative jurisdiction	Official Scheme (PMGSY, JJM, DDUGJY, 5T, SDRF, NHM)	District Collector Discretionary Fund
Text-to-Speech (TTS)	Web Speech Synthesis API + Server Base64 Audio Generator	Structured guidance prompt in citizen's active dialect	High-fidelity spoken audio waveform	Visual floating transcript dialog modal with animated equalizer

4. Complete Backend REST API Architecture (49 Endpoints)

All endpoints are served from `server.py` (Python REST server) and mirrored in `backend/server.js` (Node.js/Express).

HTTP METHOD	ENDPOINT ROUTE	PRIMARY FUNCTION	PAYLOAD / PARAMETERS
GET	/api/tts	Generates streamed or synthesized speech audio	?text=...&lang=...
GET	/api/grievances/nearby	Haversine geo-spatial query for issues within 5–10km radius	?lat=...&lng=...&radius=...
GET	/api/geo/location	Server-side IP telemetry lookup fallback	None
GET	/api/geo/reverse	High-precision reverse geocoding to village/ward level	?lat=...&lon=...
GET	/api/auth/me	Inspects active session cookie or bearer token	Authorization Header
GET	/api/grievances/list	Returns paginated grievances with category & status filters	?status=...&category=...

HTTP METHOD	ENDPOINT ROUTE	PRIMARY FUNCTION	PAYLOAD / PARAMETERS
GET	/api/grievances/{id}	Detailed case report, drone footage, citizen upvotes	URL parameter
GET	/api/admin/budget/overview	Real-time treasury balance, allocated vs expended ledger	None
GET	/api/drone/missions	Active, completed, and pending autonomous UAV flights	None
GET	/api/voice-notes/my-notes/{mobile}	Returns citizen's personal audio submissions	Mobile number
GET	/api/voice-notes/admin/all	Administrative officer voice queue	None
GET	/api/constituency	Gram Panchayat, Ward, and Block administrative hierarchy	None
GET	/api/projects	Active public works, sanctioned tenders, and contractors	None
GET	/api/clusters	Spatial density clustering coordinates for map overlays	None

HTTP METHOD	ENDPOINT ROUTE	PRIMARY FUNCTION	PAYLOAD / PARAMETERS
GET	/api/hotspots	Critical risk nodes and chronic infrastructure failure zones	None
GET	/api/datasets	Open Government Data (OGD) integration metrics	None
GET	/api/data-fusion/cases	Multi-sensor fusion case list with confidence scores	None
GET	/uploads/audio/{filename}	Streams stored citizen audio recordings	Filename
GET	/uploads/photos/{filename}	Serves citizen-captured photographic evidence	Filename
POST	/api/data-fusion/analyze	Runs automated verification across satellite + weather + reports	Case ID
POST	/api/auth/citizen/register	Registers new citizen with mobile & password	JSON credentials
POST	/api/auth/citizen/login	Authenticates citizen and issues session cookie	JSON credentials

HTTP METHOD	ENDPOINT ROUTE	PRIMARY FUNCTION	PAYLOAD / PARAMETERS
POST	/api/auth/citizen/send-otp	Dispatches mobile SMS OTP via Fast2SMS / Twilio	{ "mobile": "..." }
POST	/api/auth/citizen/verify-otp	Validates 6-digit OTP code	{ "mobile": "...", "otp": "..." }
POST	/api/auth/citizen/send-email-otp	Sends verification code to citizen email	{ "email": "..." }
POST	/api/auth/citizen/verify-email-otp	Validates email verification code	{ "email": "...", "otp": "..." }
POST	/api/auth/citizen/reset-password	Changes password after OTP verification	New credentials
POST	/api/auth/citizen/upload-avatar	Uploads citizen profile picture	Multipart / base64 image
POST	/api/auth/citizen/profile	Updates citizen name, village, Aadhaar hash	JSON profile data
POST	/api/auth/citizen/digilocker-fetch	Simulates verified document sync from DigiLocker API	Aadhaar number
POST	/api/auth/admin/login	Officer/collector login authentication	Executive credentials
POST	/api/auth/logout	Terminates session and clears cookies	None
POST	/api/grievances/endorse	Records citizen upvote / civic	{ "problemId": "..." }

HTTP METHOD	ENDPOINT ROUTE	PRIMARY FUNCTION	PAYLOAD / PARAMETERS
		endorsement	
POST	/api/grievances/ai-categorize	Returns real-time NLP classification & urgency score	{ "text": "...", "lang": "..." }
POST	/api/grievances/submit	Primary multi-modal submission endpoint	Text, voice, photo, GPS, user data
POST	/api/admin/drone/dispatch	Orders immediate autonomous UAV surveillance flight	{ "problemId": "...", "lat": ..., "lng": ... }
POST	/api/admin/budget/allocate	Authorizes treasury expenditure for scheme tender	{ "problemId": "...", "amount": ... }
POST	/api/translate	Live translation from dialect to English	{ "text": "...", "spokenLanguage": "..." }
POST	/api/speech-to-text	Transcribes raw WAV/WebM audio bytes into text	Binary audio or base64
POST	/api/voice-notes/upload	Uploads voice note file to server storage	Multipart audio
POST	/api/webhook/messaging	Twilio / WhatsApp incoming message webhook	Inbound payload

HTTP METHOD	ENDPOINT ROUTE	PRIMARY FUNCTION	PAYLOAD / PARAMETERS
POST	/api/optimizer/solve	Solves Knapsack optimization for state budget allocation	Budget limit & candidate list
POST	/api/adjudicate	Applies contractor breach penalty or closes dispute	{ "problemId": "...", "verdict": "..." }
POST	/api/grievances/clear-all	Test/demo reset of grievance ledger	Admin key
POST	/api/grievances/{id}/status	Updates lifecycle status (Pending, In Progress, Resolved)	{ "status": "..." }
POST	/api/voice-notes/admin/{id}/status	Updates voice note triage status	{ "status": "..." }

5. Multi-Sensor Data Fusion Architecture

To prevent corruption and "ghost works" (contractors billing for unperformed repairs), Lok Swar introduces an automated **Cross-Verification Matrix**:



Satellite Imagery Feed

Compares historical satellite baselines with current surface reflectance to verify road inundation, river overflows, and structural bridge collapses.



Autonomous UAV Drone Recon

Dispatches automated drone missions to capture 4K geotagged video and high-resolution still frames, feeding directly into administrative verification tickets.

Weather & Rain Telemetry

Cross-references IMD rain telemetry to confirm whether reported flash floods or drainage overflows match real-time precipitation in that Gram Panchayat.

Civic Endorsement Quorum

Requires minimum threshold of peer citizen upvotes from residents living within 2km, creating an immune system against fraudulent single-actor reports.

6. Knapsack Resource Optimizer

When government budgets are constrained (e.g. ₹50,00,000 available vs ₹3,20,00,000 needed), the **Lok Swar Optimizer** solves a *0/1 Multi-Criteria Knapsack Problem*:

```
Maximize:  $Z = \sum [ \text{Urgency\_Score}_i \times \text{Affected\_Population}_i \times \text{Risk\_Multiplier}_i \times X_i ]$   
Subject to:  $\sum [ \text{Cost}_i \times X_i ] \leq \text{Total\_Allocated\_Treasury\_Budget}$   
Where:  $X_i \in \{0, 1\}$  (Whether project 'i' is funded in current tranche)
```

This replaces political bias with mathematically optimal resource allocation, ensuring projects that save the most human lives and protect critical transit arteries are funded first.

7. Keypad Phone IVR Telephony System ()

Recognizing that millions of citizens own ₹800 basic feature phones (Nokia/JioPhone with physical numeric keypads and no internet), Lok Swar includes an independent toll-free IVR engine:

- **Architecture:** FastAPI asynchronous server () running on port 8001, integrated with Twilio Voice APIs and TwiML response generators.
- **DTMF Dialpad Menu:**
 -  → ଓଡ଼ିଆ (Odia)
 -  → हिन्दी (Hindi)
 -  → भोजपुरी (Bhojpuri)
 -  → বাংলা (Bengali)
- **Voice Recording & Audio Pipeline:** The citizen speaks their grievance after the beep. Twilio records the call audio, posts the URL to  , and the AI engine automatically transcribes the audio, generates a ticket ID, and sends an SMS receipt back to their keypad phone.

8. Autonomous Drone Reconnaissance & GIS Geofencing

The `admin.html` GIS engine utilizes **Leaflet.js** with customized OpenStreetMap CartoDB dark/light tiles:

- **GeoJSON Constituency Boundaries:** Visualizes district administrative sectors (Sundargarh, Hemgir, Lephripara, Tangarpali, Bargaon) with color-coded risk envelopes.
- **Drone Flight Dispatch Simulator:** Calculates straight-line distance, battery consumption, estimated arrival time (ETA), and generates automated UAV flight waypoints.

9. Native Android Application Wrapper (`android/`)

- **Architecture:** Native Android Gradle project (`android/app/build.gradle`) featuring an optimized hardware-accelerated `WebView` shell.
- **Device Permissions Granted:**
 - `android.permission.ACCESS_FINE_LOCATION` : Satellite GPS telemetry.
 - `android.permission.RECORD_AUDIO` : High-quality microphone recording.
 - `android.permission.CAMERA` : Live photographic evidence capture.
 - `android.permission.INTERNET` : Full duplex synchronization with backend.

10. Complete Software Stack & Tool Reference

LAYER	TECHNOLOGIES USED	WHY CHOSEN
Frontend Core	React 18, Babel Standalone, Vanilla CSS3, Tailwind CSS	Zero build-step instant portability; ultra-lightweight client bundle running effortlessly on cheap ₹5,000 Android Go smartphones.
Mapping & GIS	Leaflet.js 1.9.4, CartoDB Positron & Dark Matter tiles	Open-source, no Google Maps billing quota dependencies, superior offline caching support.

LAYER	TECHNOLOGIES USED	WHY CHOSEN
Backend Services	Python 3.10+ (FastAPI, Uvicorn, standard HTTP), Node.js Express fallback	High-concurrency async I/O with native scientific Python ecosystem for NLP and optimization solvers.
Database & Storage	MongoDB Atlas, Motor async driver, Local JSON ACID atomic fallback	Dual-layer durability: cloud-scale document flexibility with zero-dependency offline local store.
Telephony & SMS	Twilio Voice API, TwiML, Fast2SMS Gateway	Enables keypad phone access across India with automated SMS status receipt generation.
Browser Web APIs	<code>SpeechRecognition</code> , <code>SpeechSynthesis</code> , <code>AudioContext</code> , <code>navigator.geolocation</code> , <code>MediaDevices</code>	Direct zero-latency access to smartphone hardware without heavy external SDK overhead.

11. Hackathon Cross-Question Defense Strategies

Q1: "How do you prevent duplicate grievances from spamming the system?"

Answer: Lok Swar uses a **Spatial-Semantic Deduplication Filter**. When a grievance is submitted, the backend runs a spatial query within a 500-meter radius (using Haversine distance) and computes word embedding cosine similarity on the category and text. If a match is found, instead of creating a duplicate ticket, the system automatically converts the submission into a *Civic Endorsement (+1 Upvote)* on the existing active ticket.

Q2: "What if there is no internet in remote rural hamlets?"

Answer: Lok Swar is built on a **Tri-Tier Connectivity Fallback**:

1. **Full Internet:** Citizen smartphone portal with live map and camera.
2. **Intermittent Internet:** LocalStorage offline queue that records grievances locally and auto-synchronizes as soon as 2G/3G connectivity is detected.
3. **Zero Internet / Keypad Phones:** Toll-free IVR telephony server (`ivr/`) where citizens dial in via standard GSM voice network, speak in their native tongue, and receive an SMS tracking ID.

Q3: "How do you handle illiterate citizens who cannot read error messages or submit forms?"

Answer: The **Every-Click Spoken Assistant**. Every element on the screen has an audio mapping in `js/citizen/voice_engine.js`. The moment an illiterate citizen taps anywhere, the Web Audio engine speaks clear instructions in their local dialect (Bhojpuri, Odia, Bengali, Hindi), accompanied by high-contrast visual icons and audio chimes.

Q4: "Is Aadhaar number storage secure and legal?"

Answer: Yes. Lok Swar strictly adheres to the Aadhaar Act guidelines. We **never store raw 12-digit Aadhaar numbers** in plaintext. All identity entries are processed through an irreversible **SHA-256 cryptographic salt**, and only the masked format (e.g. `..... 9012`) is displayed on the UI for citizen confirmation.